AOUANOUK Slimane

+33 7 83 54 30 26 — aouanoukslim@gmail.com — LinkedIn — slimaneaouanouk.com

Available for a 6-month internship starting February 2027 in robotics or autonomous systems.

Graduate Engineering student at Arts et Métiers pursuing a **Research Master's (M2) in Advanced Systems and Robotics at Sorbonne Université**. Research experience in **visual localization, motion planning, and real-time mobile robotics**. Currently developing a **Vision-Language-Action pipeline for Unitree G1 assistive manipulation**.

EDUCATION — Combined CPGE–ENSAM GPA: 3.92 / 4.00

Sorbonne Université — Paris, France **Sept 2026 – Aug 2027**
Research Master's (M2) in Advanced Systems & Robotics (SAR)
Current coursework: Mobile Robotics (ROS 2, SLAM), Robot Perception, State Estimation & System Identification (Bayesian estimation, EKF, UKF, particle filters), Machine Learning (deep learning, reinforcement learning, PyTorch), Nonlinear Control, and Robot Manipulators.

Arts et Métiers Institute of Technology (ENSAM) — Paris, France **Sept 2024 – Jul 2027**
Combined BS × MS Engineering Degree — *Member of* ***ParisTech***.
- **Ranked top 12% of cohort** (**142 / 1223** students)
- **Relevant Coursework:** Robotics and Mechatronics, Numerical Optimization, Mechanical Design, Manufacturing, and Experimental Mechanics

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems **Jan–Aug 2026**
Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisor: Prof. Francisco Chinesta
- **Co-developed a Python/OpenCV visual-localization pipeline** using a Basler overhead camera and ArUco markers to estimate rover position and heading for closed-loop mobile-robot navigation.
- **Calibrated camera intrinsics using 20 ChArUco views**, achieving **0.191 px reprojection RMS**; combined lens-distortion correction with metric homography to reduce mean localization error **84%, from 36.3 to 5.8 mm**, on an independent nine-point ground grid.
- **Diagnosed and corrected height-induced parallax**, reducing mean rover-marker localization error from **60.8 to 11.6 mm** across five physical validation positions.
- **Profiled and optimized vision processing through image downsampling and point-wise distortion correction**, integrating localization, trajectory correction, serial rover control, and experiment logging at a median **19.84–19.86 Hz**.
- **Validated the integrated navigation stack through two frozen 20-run physical campaigns**, achieving **40/40 goal arrivals with zero collisions** and collecting synchronized control and trajectory logs.
- **Evaluated trajectory-planning and control configurations through a 100,800-case simulation campaign** and quantified simulation-to-real discrepancies across **39 paired physical and digital-twin runs**.
- **Co-developed pytest unit and regression tests** covering calibration, homography, controller behavior, and analytical gradients, including finite-difference checks for numerical consistency.
- **First author of a robotics manuscript currently under journal review**, presenting the trajectory-planning framework, experimental validation, and simulation-to-real evaluation.

SELECTED PROJECTS

Humanoid Robot Learning for Assistive Manipulation — VLA **2026–27**
Sorbonne Université · Supervised by Prof. Olivier Bruneau, LURPA / ENS Paris-Saclay
- **Developing an end-to-end Vision-Language-Action (VLA) pipeline** for autonomous manipulation on a Unitree G1 humanoid, from teleoperated demonstrations to real-world deployment.
- **Targeting OpenPI (π_0/$\pi_{0.5}$) policy fine-tuning** using imitation learning and multimodal datasets combining natural-language instructions, RGB/RGB-D observations, and robot actions.
- **Exploring language-conditioned manipulation policies** for household assistance tasks, targeting robust execution, generalization to unseen configurations, and quantitative real-world evaluation.
- **Exploring AI-based motion optimization in MuJoCo or NVIDIA Isaac Lab (Isaac Sim)**, with motion capture and human-to-robot retargeting envisaged to adapt human movements to the robot.

TIPE — Experimental Impact Testing & Python Video Analysis **2022–24**
- Individually built a **46 cm pendulum-style impact rig with four contact geometries** for repeatable tests.
- **Analyzed impact videos in Python (NumPy, SciPy, Matplotlib)** using pre/post-impact regressions and position-uncertainty plots; measured **4.0 m/s ball exit speed from a 2.2 m/s impactor** (12 mm cylinder, 3 psi).

Foundry, Machining and Metrology – Arts et Métiers **2025**
- Co-manufactured an AlSi7 bearing-support part through casting, turning, and CNC machining; **inspected four geometric tolerances on a CMM**, documenting two nonconformities and tracing setup and datum errors for redesign.

SKILLS AND INTERESTS

Programming & Numerics: Python, NumPy, SciPy, MATLAB, pytest (unit/regression tests), Git, LaTeX
Robotics & Vision: OpenCV, ArUco localization, camera calibration, motion planning, real-time robot integration
Current Coursework: ROS 2, SLAM, Bayesian state estimation, PyTorch, Linux
Engineering Tools: CAD (Fusion 360, 3DEXPERIENCE), Simulink, CMM metrology, digital twins
Languages: French (native), English (TOEFL iBT: 5/6), Italian (basic), Spanish (basic)
Interests: Autonomous navigation, robot learning, physical AI, basketball, strength training